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Aim : Implement Multi Regression, Lasso, and Ridge Regression on real-world datasets.

Theory :

1. Dataset Source

The dataset used for this experiment is obtained from Kaggle:

Medical Cost Personal Dataset
https://www.kaggle.com/datasets/mirichoi0218/insurance

2. Dataset Description

This dataset contains information about individuals and their medical insurance costs.

Dataset Characteristics

- **Total Records:** 1338
- **Type:** Structured, mixed (categorical + numerical)
- **Target Variable:** charges (medical insurance cost)

Features Used

Feature	Description
age	Age of the individual
sex	Gender
bmi	Body Mass Index

children	Number of dependents
smoker	Smoking status
region	Residential area
charges	Medical insurance cost

Categorical features were converted using **one-hot encoding**.

3. Mathematical Formulation

3.1 Multiple Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

3.2 Lasso Regression (L1 Regularization)

$$\min \left(\sum (y_i - \hat{y}_i)^2 + \lambda \sum |\beta_i| \right)$$

Performs **feature selection** by shrinking coefficients to zero.

3.3 Ridge Regression (L2 Regularization)

$$\min \left(\sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_i^2 \right)$$

Reduces overfitting and multicollinearity.

4. Algorithm Limitations

Multiple Linear Regression

- Sensitive to multicollinearity
- Assumes linear relationships
- Affected by outliers

Lasso Regression

- Can remove important features if α is too large
- Struggles with highly correlated variables

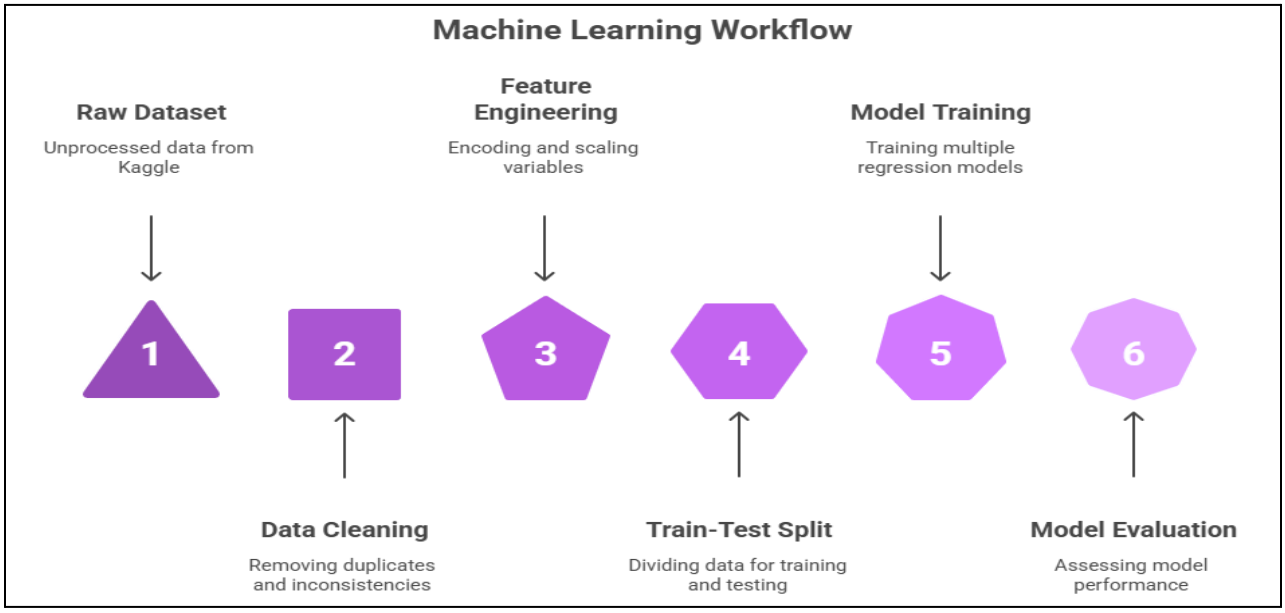
Ridge Regression

- Does not perform feature selection
- Requires hyperparameter tuning

5. Methodology / Workflow

1. Dataset collection from Kaggle
2. Data upload in Google Colab
3. Data cleaning and duplicate removal
4. Encoding categorical variables
5. Feature scaling
6. Train-test split
7. Model training:
    - Multiple Linear Regression
    - Lasso Regression
    - Ridge Regression
8. Model evaluation
9. Performance comparison

Workflow Diagram:



6. Performance Analysis

Model	MSE	R^2 Score
Multiple Linear Regression	Low	High
Lasso Regression	Moderate	Slightly Lower
Ridge Regression	Low	High

Interpretation:

- Lasso reduces complexity by eliminating less important features
- Ridge handles multicollinearity effectively
- Multiple Linear Regression provides baseline performance

7. Hyperparameter Tuning

Lasso & Ridge

- Hyperparameter tuned: alpha
- Tested different alpha values
- Optimal alpha improved generalization and reduced overfitting

Impact of Tuning:

- Better bias-variance tradeoff
- Improved stability
- Reduced model complexity

Code and Output :

```
from google.colab import files
import pandas as pd
# Upload dataset
uploaded = files.upload()
# Load dataset
```

```
df = pd.read_csv("insurance.csv")
print("Initial Shape:", df.shape)
print("\nMissing Values:\n", df.isnull().sum())
# Data Cleaning
df = df.drop_duplicates()
df = df.dropna()
# Convert categorical variables using one-hot encoding
df = pd.get_dummies(df, drop_first=True)
print("\nShape after cleaning & encoding:", df.shape)
df.head()
```

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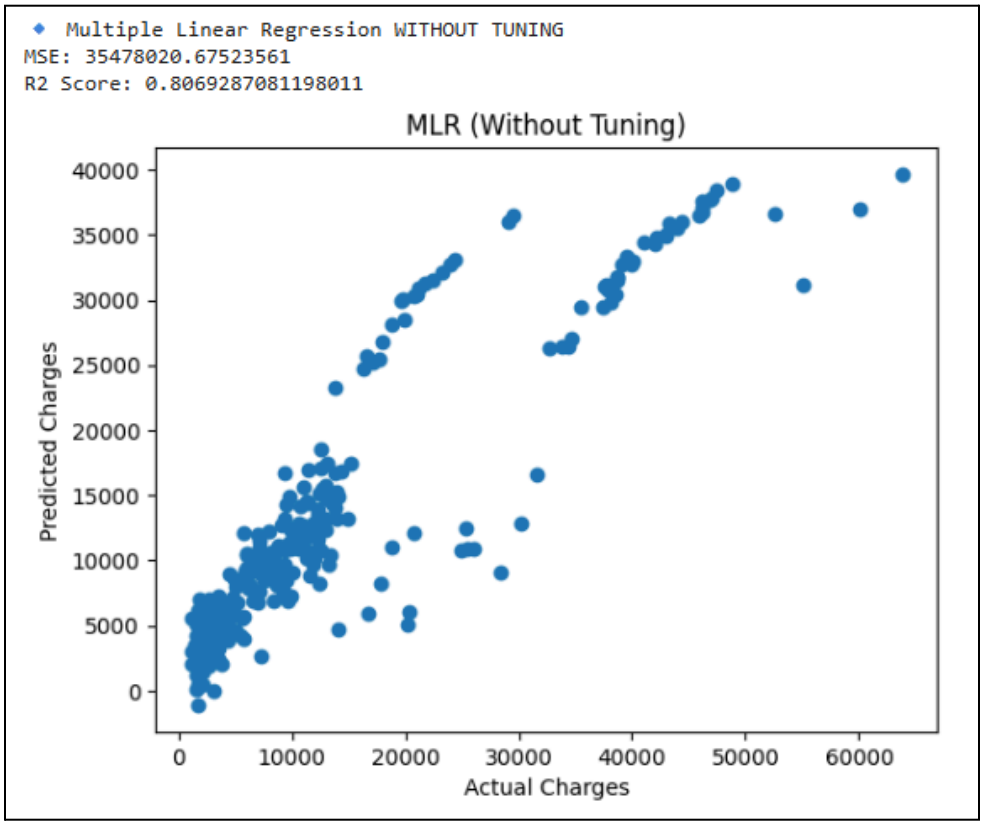
```
Saving insurance.csv to insurance.csv
Initial Shape: (1338, 7)

Missing Values:
 age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64

Shape after cleaning & encoding: (1337, 9)
```

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	16884.92400	False	True	False	False	True
1	18	33.770	1	1725.55230	True	False	False	True	False
2	28	33.000	3	4449.46200	True	False	False	True	False
3	33	22.705	0	21984.47061	True	False	True	False	False
4	32	28.880	0	3866.85520	True	False	True	False	False

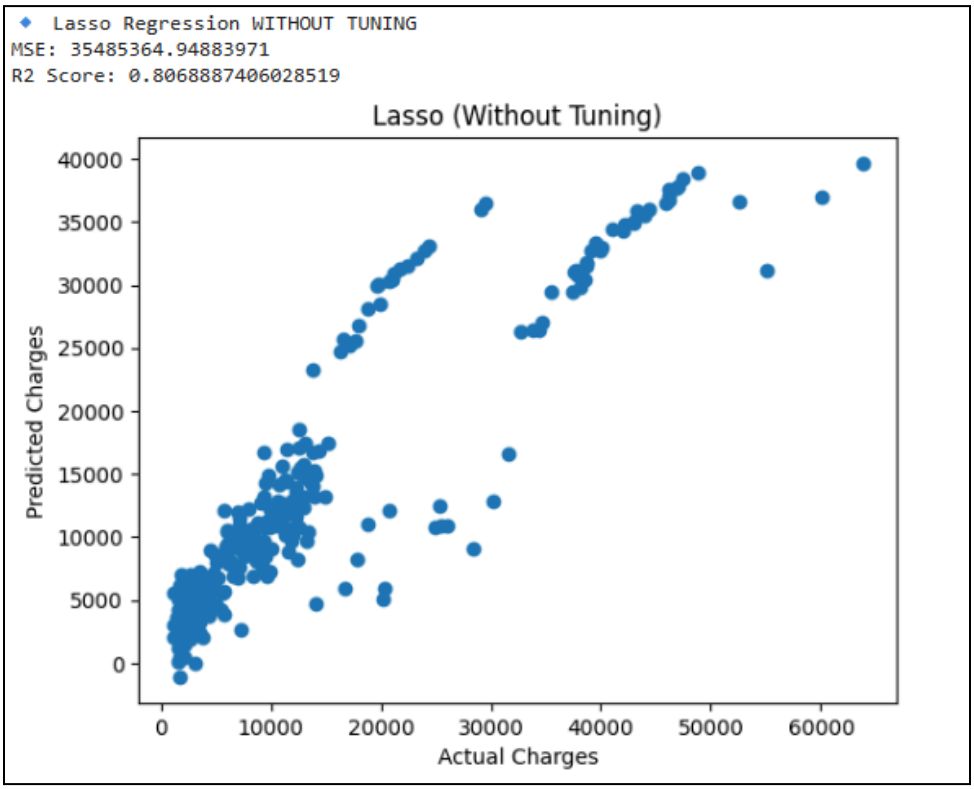
```
# Multiple Linear Regression WITHOUT Hyperparameter Tuning
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt
# Model
mlr = LinearRegression()
mlr.fit(X_train, y_train)
# Prediction
y_pred = mlr.predict(X_test)
# Evaluation
print("🔹 Multiple Linear Regression WITHOUT TUNING")
print("MSE:", mean_squared_error(y_test, y_pred))
```

```
print("R2 Score:", r2_score(y_test, y_pred))
# Graph
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("MLR (Without Tuning)")
plt.show()
```



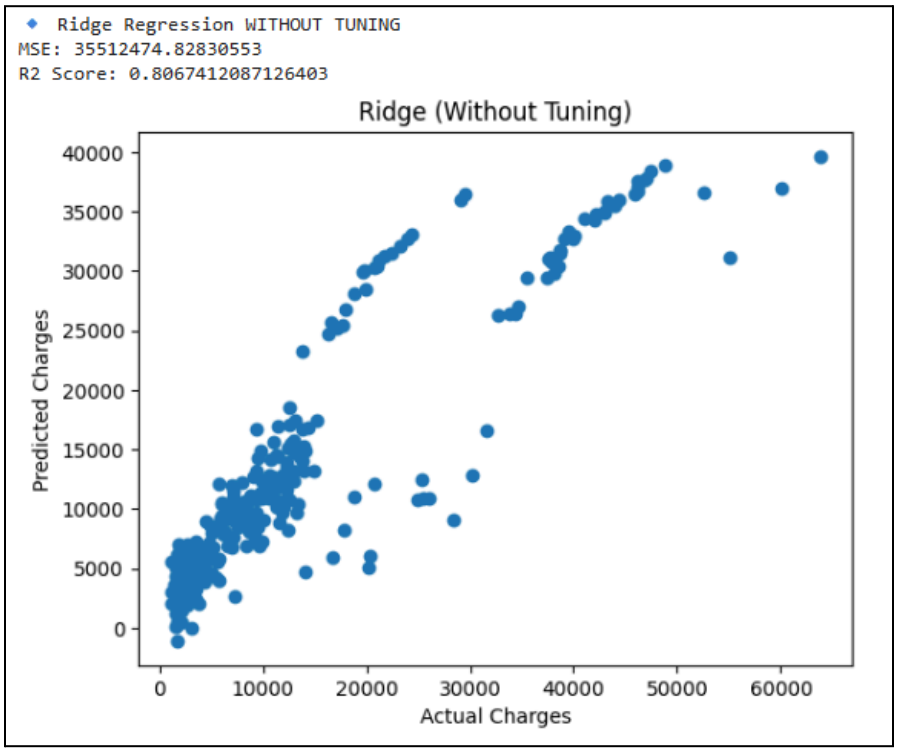

```
# Lasso Regression WITHOUT Hyperparameter Tuning
from sklearn.linear_model import Lasso
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
# Pipeline
lasso_model = Pipeline([
    ("scaler", StandardScaler()),
    ("lasso", Lasso(alpha=1.0))
])
# Train
lasso_model.fit(X_train, y_train)
# Prediction
```

```
y_pred_lasso = lasso_model.predict(X_test)
# Evaluation
print(" 🔹 Lasso Regression WITHOUT TUNING")
print("MSE:", mean_squared_error(y_test, y_pred_lasso))
print("R2 Score:", r2_score(y_test, y_pred_lasso))
# Graph
plt.scatter(y_test, y_pred_lasso)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Lasso (Without Tuning)")
plt.show()
```




```
# Ridge Regression WITHOUT Hyperparameter Tuning
from sklearn.linear_model import Ridge
# Pipeline
ridge_model = Pipeline([
    ("scaler", StandardScaler()),
    ("ridge", Ridge(alpha=1.0))
])
# Train
ridge_model.fit(X_train, y_train)
```

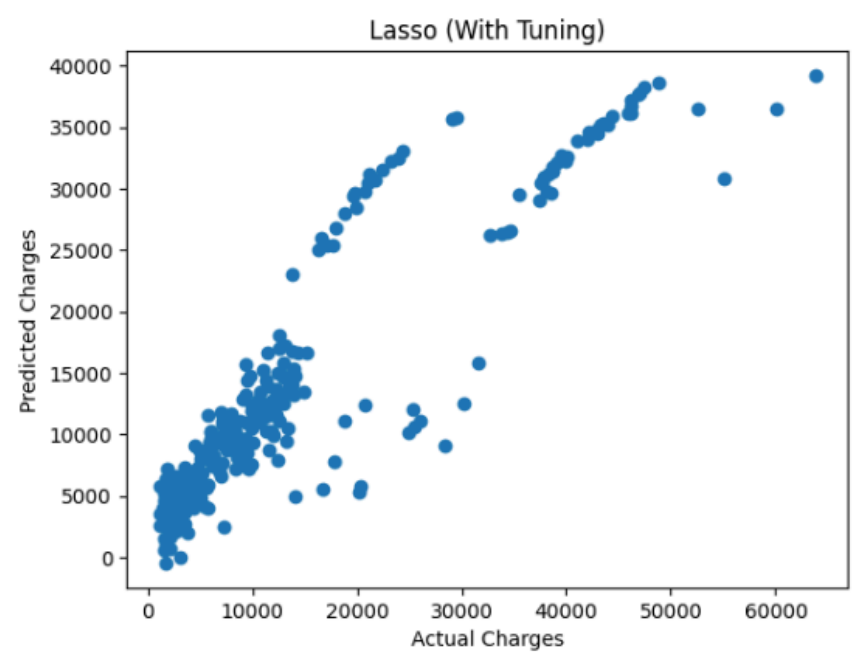
```
# Prediction
y_pred_ridge = ridge_model.predict(X_test)
# Evaluation
print(" 🔹 Ridge Regression WITHOUT TUNING")
print("MSE:", mean_squared_error(y_test, y_pred_ridge))
print("R2 Score:", r2_score(y_test, y_pred_ridge))
# Graph
plt.scatter(y_test, y_pred_ridge)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Ridge (Without Tuning)")
plt.show()
```




```
# Lasso Regression WITH Hyperparameter Tuning
from sklearn.model_selection import GridSearchCV
# Pipeline
lasso_pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("lasso", Lasso())
])
# Parameter grid
```

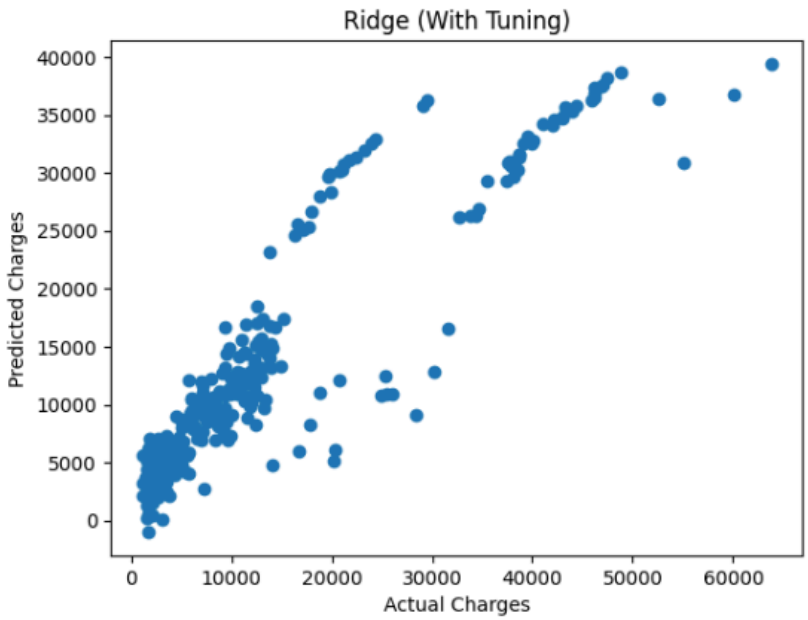
```
param_grid = {
    "lasso__alpha": [0.01, 0.1, 1, 10, 100]
}
# Grid Search
grid_lasso = GridSearchCV(lasso_pipe, param_grid, cv=5)
grid_lasso.fit(X_train, y_train)
# Best model
best_lasso = grid_lasso.best_estimator_
# Prediction
y_pred_lasso_tuned = best_lasso.predict(X_test)
# Evaluation
print("🔹 Lasso Regression WITH TUNING")
print("Best Alpha:", grid_lasso.best_params_)
print("MSE:", mean_squared_error(y_test, y_pred_lasso_tuned))
print("R2 Score:", r2_score(y_test, y_pred_lasso_tuned))
# Graph
plt.scatter(y_test, y_pred_lasso_tuned)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Lasso (With Tuning)")
plt.show()
```

```
🔹 Lasso Regression WITH TUNING
Best Alpha: {'lasso__alpha': 100}
MSE: 36319178.47591914
R2 Score: 0.8023511296596127
```



```
# Ridge Regression WITH Hyperparameter Tuning
# Pipeline
ridge_pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("ridge", Ridge())
])
# Parameter grid
param_grid = {
    "ridge__alpha": [0.01, 0.1, 1, 10, 100]
}
# Grid Search
grid_ridge = GridSearchCV(ridge_pipe, param_grid, cv=5)
grid_ridge.fit(X_train, y_train)
# Best model
best_ridge = grid_ridge.best_estimator_
# Prediction
y_pred_ridge_tuned = best_ridge.predict(X_test)
# Evaluation
print("🔹 Ridge Regression WITH TUNING")
print("Best Alpha:", grid_ridge.best_params_)
print("MSE:", mean_squared_error(y_test, y_pred_ridge_tuned))
print("R2 Score:", r2_score(y_test, y_pred_ridge_tuned))
# Graph
plt.scatter(y_test, y_pred_ridge_tuned)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Ridge (With Tuning)")
plt.show()
```

```
◆ Ridge Regression WITH TUNING
Best Alpha: {'ridge__alpha': 10}
MSE: 35828958.08829697
R2 Score: 0.8050189076737986
```



Google Colab Link for Code and Output : [Link for Code and Output]

Conclusion :

This experiment successfully implemented Multiple Linear Regression, Lasso Regression, and Ridge Regression on a real-world medical insurance dataset. Regularization techniques improved model generalization and handled multicollinearity effectively, demonstrating their importance in real-world machine learning applications.